STROKE REHABILITATION MONITORING SYSTEM

Comprehensive Research Corpus & Literature Review

For ECG, Heart Rate, and Movement Signal Analysis in Stroke Recovery

Target Population Adults 40–65+ Inpatient & outpatient	**Data Modalities** ECG, Heart Rate, HRV IMU / Movement signals
System Layers Wearable → ML → LLM → Agent	**Research Pillars** 5 domains, 20 sub-categories

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Table of Contents

Table of Contents 2
Executive Summary 4
1. Stroke Physiology & Cardiac Interaction 5
1.1 Cardioembolic Stroke Mechanisms 5
Priority paper categories for A1 5
1.2 Age-Stratified Cardiac Risk Profiles 5
Key ECG and HR biomarkers by age group 6
1.3 Post-Stroke Autonomic Dysfunction 6
Priority paper categories for A3 7
1.4 Comorbidity Interaction Papers 7
2. Recovery Staging & Clinical Benchmarks 8
2.1 Validated Clinical Recovery Scales 8
2.2 Temporal Recovery Trajectories 9
Recovery phase characteristics by signal domain 9
Priority sources for B2 9
2.3 Neuroplasticity & Motor Recovery 9
2.4 Inpatient vs. Outpatient Rehabilitation Differences 10
3. Wearable Signal Processing & ECG Analysis 11
3.1 Heart Rate Variability as a Stroke Recovery Biomarker 11
HRV metrics and their clinical interpretation 11
3.2 ECG-Based Arrhythmia Detection Post-Stroke 12
Arrhythmia detection performance benchmarks 12
3.3 Ambulatory Movement Analysis in Neurorehabilitation 12
3.4 Signal Quality & Artifact Rejection 13
Key artifact sources by signal type 13
Priority methods papers for C4 13
3.5 Sensor Fusion Methods 13
4. Data Correlation & Clinical Benchmark Mapping 15
4.1 Wearable-to-Clinical Benchmark Validation Studies 15
4.2 Digital Biomarker Discovery 15
Emerging digital biomarkers in stroke rehabilitation 15
4.3 Remote Patient Monitoring Clinical Trials 16
4.4 Physiological Signal Databases for Training & Validation 16

5. Clinical AI Architecture & Agent Design....18
5.1 Clinical NLP & Medical LLM Fine-Tuning....18
Key medical LLM architectures and their validation....18
5.2 Explainability in Clinical AI....18
5.3 AI Safety & Alert Fatigue....19
5.4 Human-AI Collaboration in Rehabilitation....19
For the clinician interface....20
For the patient interface....20
6. Corpus Curation Methodology....21
6.1 Search Strategy....21
6.2 Inclusion & Exclusion Criteria....21
Inclusion criteria....21
Exclusion criteria....21
6.3 Tagging Schema for LLM Ingestion....21
6.4 Recommended Corpus Size & Composition....22
7. Conclusions & Next Steps....23
Recommended next steps....23

Executive Summary

This literature review establishes the research corpus required to design, train, and validate a Stroke Rehabilitation Monitoring System (SRMS). The system integrates wearable sensor data — electrocardiogram (ECG), heart rate (HR), and movement signals — with machine learning, a large language model (LLM), and a clinical AI agent to support both patients and clinicians across the full arc of stroke recovery.

The corpus is organized across five research pillars, each feeding one or more of the system's four architectural layers. Together they address the three core research requirements identified in the project brief: the role of cardiac conditions and age in stroke scenarios; data-driven descriptions of recovery trajectories; and validation of wearable sensor data against established clinical benchmarks.

Scope note

This review covers adult stroke patients aged 40 and above, including middle-aged adults (40–64) and older adults (65+), across both post-acute inpatient rehabilitation and outpatient or home-based settings. Where evidence differs meaningfully by age group or setting, this is made explicit.

Research Pillar	Primary Focus	System Layer Fed
A — Stroke physiology & cardiac	Cardiac mechanisms, autonomic function, age-related risk	ML pipeline, LLM, Agent
B — Recovery staging & benchmarks	Clinical scales, recovery trajectories, neuroplasticity	LLM + clinical fusion, Agent
C — Wearable signal processing	ECG features, HRV, movement analysis, sensor fusion	Wearable collection, ML pipeline
D — Data correlation & mapping	Wearable-to-clinical validation, digital biomarkers, datasets	LLM + clinical fusion
E — Clinical AI architecture	LLM fine-tuning, explainability, safety, HCI	LLM + clinical fusion, Agent

Pillar A — Stroke Physiology & Cardiac Interaction

1. Stroke Physiology & Cardiac Interaction

Understanding the bidirectional relationship between cardiac function and stroke is the foundational requirement for every downstream component of the SRMS. This pillar spans four sub-domains: the mechanisms by which cardiac pathology causes stroke; how these risks vary with age; how the stroke event itself disrupts cardiac autonomic control; and how comorbidities compound both risk and recovery.

1.1 Cardioembolic Stroke Mechanisms

Cardioembolic stroke accounts for approximately 20–25% of all ischemic strokes, making it the single most clinically important mechanism for the SRMS to recognise. Atrial fibrillation (AF) is the dominant cause, responsible for 15–20% of ischemic strokes. AF produces thrombus formation in the left atrial appendage, from which emboli travel to cerebral vessels. The 12-lead ECG and ambulatory monitoring are the primary detection instruments.

Other cardiac sources include patent foramen ovale (PFO) — increasingly implicated in cryptogenic stroke in patients under 60 — dilated cardiomyopathy, mechanical heart valves, and infective endocarditis. Research in this sub-domain should be prioritized particularly for the ML pipeline's arrhythmia detection module and for the AI agent's cardiac risk reasoning.

Key journals

Stroke (AHA); Circulation; Journal of the American College of Cardiology (JACC); European Heart Journal; Cerebrovascular Diseases

Priority paper categories for A1

- RCTs and meta-analyses on AF screening strategies post-stroke (e.g., Holter vs. patch monitor duration)
- Cohort studies quantifying embolic risk by cardiac etiology and CHA_2DS_2-VASc score
- ECG feature studies identifying P-wave morphology, PR interval, and QRS changes predictive of AF
- Comparative studies of cardioembolic stroke phenotypes across age groups (40–64 vs. 65+)
- Reviews on secondary prevention pharmacology relevant to wearable-triggered alert thresholds

1.2 Age-Stratified Cardiac Risk Profiles

Age is the single most powerful non-modifiable risk factor for both stroke incidence and post-stroke cardiac complications. The SRMS must be calibrated to generate different alert thresholds and clinical interpretations depending on the patient's age group, as the physiological baseline and expected deviation ranges differ significantly.

In middle-aged adults (40–64), AF is less prevalent but coronary artery disease, hypertension, and metabolic syndrome dominate the cardiac risk profile. In older adults (65+), AF prevalence rises sharply (to approximately 10% in those over 80), heart rate variability (HRV) is intrinsically lower due to reduced sinoatrial node function, and orthostatic hypotension is more common, complicating the interpretation of resting HR and HRV features.

Key ECG and HR biomarkers by age group

Biomarker	Middle-aged (40–64)	Older adults (65+)
Resting HR (bpm)	60–80 typical	Slightly elevated; rate control drugs common
SDNN (HRV, ms)	>50ms normal	Lower baselines; <30ms more common
QTc interval	Normal <450ms (M), <460ms (F)	Prolongation risk elevated; drug interactions
PR interval	120–200ms	First-degree AV block more common
AF prevalence	~2–3% population	~5–10% population; paroxysmal AF common

ML calibration implication

Any feature threshold used to classify 'abnormal' HRV or resting HR must be stratified by age group. A SDNN of 35ms is clinically concerning in a 48-year-old but may be within normal limits for a 78-year-old on beta-blockers. The corpus should include normative reference data from large epidemiological studies such as the Framingham Heart Study and UK Biobank.

1.3 Post-Stroke Autonomic Dysfunction

The stroke event itself disrupts the central autonomic network (CAN), particularly when lesions involve the insular cortex, hypothalamus, or brainstem. This produces a well-characterised syndrome of post-stroke autonomic dysfunction: elevated sympathetic tone, reduced parasympathetic activity, and blunted baroreflex sensitivity. These changes are clinically significant for three reasons.

First, they directly alter the ECG and HR signals that the SRMS collects, producing elevated resting HR, reduced HRV, and QT prolongation in the acute and subacute phases. This means

that ‘normal’ post-stroke wearable readings are categorically different from healthy baselines. Second, autonomic dysfunction predicts secondary cardiac events: neurogenic cardiac injury, Takotsubo syndrome, and sudden cardiac death are all more common after stroke. Third, recovery of autonomic function is a meaningful clinical endpoint that the wearable system can track.

Priority paper categories for A3

- Studies mapping lesion location to degree of autonomic dysfunction (insular stroke, brainstem stroke)
- Longitudinal HRV studies in the first 7–30 days post-stroke establishing post-stroke normal ranges
- Papers on Neurogenic Stunned Myocardium (NSM) and Takotsubo cardiomyopathy as stroke complications
- Baroreflex sensitivity studies as predictor of 30-day and 12-month cardiac mortality
- Papers examining autonomic recovery as a biomarker of neurological recovery over 3–12 months

1.4 Comorbidity Interaction Papers

In clinical practice, stroke patients rarely present with isolated risk factors. The SRMS’s AI agent must be capable of conditional reasoning — adjusting its interpretation of wearable signals based on the patient’s comorbidity profile. The most clinically important interactions are between AF, hypertension, type 2 diabetes, and heart failure.

Hypertension directly affects HRV and resting HR through structural cardiac remodelling (left ventricular hypertrophy) and arterial stiffness. Type 2 diabetes causes diabetic autonomic neuropathy, which reduces HRV and predisposes to cardiac arrhythmias independently of stroke. Heart failure reduces exercise tolerance and produces distinctive HR patterns (elevated resting HR, blunted rate response to activity) that must be distinguished from stroke-related autonomic dysfunction.

- Meta-analyses quantifying comorbidity burden and post-stroke mortality (NIHSS scores stratified by comorbidity)
- Studies on diabetic cardiac autonomic neuropathy and its interaction with post-stroke HRV changes
- HFpEF and HFrEF-specific cardiac monitoring studies — distinguishing heart failure signatures in wearable HR data
- Polypharmacy studies: how beta-blockers, rate-control drugs, and antihypertensives shift ECG baselines

Pillar B — Recovery Staging & Clinical Benchmarks

2. Recovery Staging & Clinical Benchmarks

The SRMS cannot learn what recovery looks like without a thorough grounding in validated clinical scales, longitudinal recovery data, and the neuroscience of motor rehabilitation. This pillar provides the ground-truth labels from which the ML model learns and the LLM reasons.

2.1 Validated Clinical Recovery Scales

Stroke recovery is assessed using a set of internationally validated instruments, each measuring a different dimension of functional outcome. The corpus must include both the foundational validation studies for each instrument and contemporary studies correlating instrument scores with physiological signals detectable by wearable sensors.

Scale	What it Measures	Relevance to Wearable Data
NIHSS	Acute neurological deficit (0–42)	Baseline severity; predicts recovery trajectory shape
Modified Rankin (mRS)	Global disability (0–6)	Gold-standard outcome label for ML training at 90 days
Barthel Index	Activities of daily living (0–100)	Correlates with step count, limb use asymmetry from IMU
Fugl-Meyer Assessment	Limb motor function (0–226)	Upper limb subscore maps to wrist accelerometry data
Berg Balance Scale	Balance and fall risk (0–56)	Sit-to-stand transitions detectable via IMU
FIM (Functional Independence)	Functional independence (18–126)	Composite; activity-based IMU features correlate

LLM training implication

Each paper in this sub-domain should be tagged with the scale it reports, the time point of assessment (acute / subacute / chronic), and whether it includes any physiological signal correlates. This metadata enables the RAG system to retrieve the most relevant benchmark reference when the AI agent is reasoning about a specific patient's recovery stage.

2.2 Temporal Recovery Trajectories

Recovery from stroke is not linear and is not uniform. The literature distinguishes three phases: the acute phase (day 1–7), characterised by rapid spontaneous recovery driven by resolution of oedema and reperfusion; the subacute phase (weeks 1–12), during which rehabilitation-driven neuroplasticity dominates; and the chronic phase (months 3–12+), in which recovery slows significantly.

Large registry studies provide the most reliable trajectory data. The NINDS Stroke Data Bank, the Copenhagen Stroke Study, and the Framingham Heart Study longitudinal cohort are primary sources. More recently, real-world data from electronic health records (EHRs) of large hospital networks have provided trajectory data at scale.

Recovery phase characteristics by signal domain

Phase	HR/HRV Pattern	Movement Pattern	Clinical Correlates
Acute (day 1–7)	Elevated HR; very low HRV; QT prolonged	Minimal purposeful movement; bed-bound	NIHSS assessment; medical stabilisation
Subacute (wk 1–12)	Gradual HR normalisation; HRV recovery begins	Progressive mobility; PT-driven activity	Barthel, FMA assessments; inpatient rehab
Chronic (mo 3–12+)	Near-baseline HR; HRV remains below pre-stroke	Asymmetric gait; compensatory movements	mRS, community reintegration metrics

Priority sources for B2

- Copenhagen Stroke Study (Jorgensen et al.) — Stroke, 1995–2000 series: gold-standard natural history data
- NINDS rt-PA Stroke Study Group datasets: acute physiological trajectories with 90-day mRS outcomes
- Virtual International Stroke Trials Archive (VISTA): harmonised individual patient data from multiple RCTs
- UK Biobank accelerometry linkage studies: population-level movement recovery benchmarks
- EHR-based trajectory studies from large academic medical centres (UCSF, Mayo Clinic data publications)

2.3 Neuroplasticity & Motor Recovery

The biological basis of motor recovery is cortical and subcortical reorganisation — the brain's capacity to recruit alternative neural circuits to compensate for damaged tissue. Understanding

this mechanism is essential for the AI agent to distinguish genuine motor recovery (new neural pathways) from compensation (using unaffected limbs to bypass the deficit).

The wearable IMU data stream is particularly valuable here. Studies using wrist accelerometry and gyroscopes have demonstrated that the ratio of affected-to-unaffected limb use (known as the Activity Ratio) correlates strongly with Fugl-Meyer upper extremity scores and with functional MRI measures of cortical reorganisation. This provides a direct bridge between raw sensor data and neurological recovery.

- Constraint-Induced Movement Therapy (CIMT) trials measuring limb use ratio with accelerometry (Taub et al.)
- Longitudinal fMRI-accelerometry correlation studies (e.g., work from the UCSF Stroke Recovery Lab)
- Papers on bilateral movement symmetry as predictor of functional outcome at 6 months
- Gait analysis studies correlating step count, cadence, and stride length with Barthel scores
- Sit-to-stand transition studies using IMU — clinically validated against Berg Balance Scale

2.4 Inpatient vs. Outpatient Rehabilitation Differences

The clinical context in which rehabilitation occurs profoundly affects the interpretation of wearable data. Inpatient rehabilitation settings provide structured, supervised therapy with continuous monitoring, while home-based outpatient rehabilitation exposes patients to unstructured real-world environments with greater autonomy and greater safety risk.

For the AI agent, this distinction is critical. A low activity level reading in an inpatient setting on day 3 post-stroke is expected. The same reading at week 8 in an outpatient setting may indicate a deterioration or non-adherence requiring clinical attention. The corpus should include comparative effectiveness research on rehabilitation setting and intensity to calibrate these contextual thresholds.

- Post-Acute Stroke Services (PASS) trial data comparing inpatient and home-based outcomes
- Telerehabilitation RCTs with physiological and activity monitoring (e.g., VStroke trial)
- Studies on rehabilitation intensity dose-response: hours of therapy per week vs. functional outcome
- Home-based fall risk studies in stroke survivors with IMU-based detection

Pillar C — Wearable Signal Processing & ECG Analysis

3. Wearable Signal Processing & ECG Analysis

This is the most technically intensive pillar and feeds directly into the ML pipeline (Layer 2) and the wearable integration layer (Layer 1). It spans five sub-domains: HRV as a clinical biomarker; ECG-based arrhythmia detection; movement analysis; signal quality and artifact rejection; and sensor fusion methods.

3.1 Heart Rate Variability as a Stroke Recovery Biomarker

Heart rate variability — the beat-to-beat variation in the RR interval — is a non-invasive window into autonomic nervous system function. It is one of the most well-studied wearable biomarkers in cardiovascular and neurological research, and its reduction post-stroke is one of the most reproducible physiological findings in the literature.

HRV metrics and their clinical interpretation

HRV Metric	Domain	Clinical Meaning in Stroke
SDNN	Time-domain	Overall autonomic function; <30ms poor prognosis
RMSSD	Time-domain	Parasympathetic tone; blunted in acute stroke
pNN50	Time-domain	Vagal activity; correlates with consciousness level
LF/HF ratio	Frequency-domain	Sympathovagal balance; elevated in acute stroke
DFA α1	Nonlinear	Fractal correlation; predicts AF in ambulatory ECG

- Systematic reviews of HRV as predictor of post-stroke functional outcome (search: ‘HRV stroke prognosis meta-analysis’)
- Studies establishing HRV recovery curves over 90 days — essential for temporal threshold calibration
- Papers comparing HRV derived from chest-electrode ECG vs. wrist PPG — critical for device selection
- HRV normative data stratified by age, sex, and fitness level (Shaffer & Ginsberg, 2017, Frontiers in Public Health)

- Studies on nocturnal HRV as predictor of daytime autonomic recovery

3.2 ECG-Based Arrhythmia Detection Post-Stroke

Continuous or intermittent ECG monitoring is the most clinically urgent technical capability of the SRMS. Post-stroke AF detection in particular is a high-stakes task: AF detected after stroke triggers anticoagulation decisions that significantly reduce secondary stroke risk (30–60% relative risk reduction with appropriate therapy).

The literature on wearable ECG for AF detection has expanded rapidly with the proliferation of patch monitors, smartwatches with single-lead ECG, and AI-based detection algorithms. The AliveCor KardiaMobile, Apple Watch ECG, and Zio patch (iRhythm) are the most studied commercial devices and their validation studies are essential references.

Arrhythmia detection performance benchmarks

Device / Method	AF Sensitivity	AF Specificity	Key Reference Type
12-lead hospital ECG	~60–70% (paroxysmal AF)	~99%	Standard of care baseline
24-hour Holter monitor	~70–80%	~97%	Outpatient standard
30-day event monitor	~85–90%	~95%	Post-stroke guideline recommendation
Zio patch (14–30 days)	~90–95%	~97%	RCT validation data
Apple Watch ECG (single-lead)	~84%	~95%	Consumer-grade; validation studies available
Deep learning on wearable ECG	~95%+	~95%+	Algorithm papers; PhysioNet challenge results

Clinical safety note

For the AI agent's alert logic, sensitivity is the paramount concern — a missed AF detection that leads to an unprotected secondary stroke is a catastrophic outcome. The corpus should include studies on the clinical consequences of false negatives in this context, alongside papers on the regulatory pathway for AI-based arrhythmia detection (FDA De Novo and 510(k) submissions).

3.3 Ambulatory Movement Analysis in Neurorehabilitation

The IMU (inertial measurement unit) data stream — combining accelerometry and gyroscopy — provides the most direct measure of motor recovery in the home environment. The literature in

this domain spans from basic biomechanical studies establishing the validity of IMU-derived gait parameters to clinical trials using wearable activity monitors as primary outcome measures.

- Upper limb: Accelerometry-based limb use ratio studies (affected vs. unaffected); Fugl-Meyer correlation
- Gait: Step count, cadence, stride length, and gait asymmetry from bilateral ankle IMUs
- Balance: Sit-to-stand transitions, sway measures during standing — Berg Balance Scale correlation studies
- Activity counts: Raw accelerometry-based activity counts vs. Barthel Index and FAC (Functional Ambulation Categories)
- Real-world vs. lab performance discrepancy studies — critical for home-based monitoring calibration
- Sedentary behaviour duration as predictor of 6-month recovery — emerging literature (2019–2024)

3.4 Signal Quality & Artifact Rejection

The clinical utility of the SRMS depends entirely on the quality of the signals it collects. In real-world rehabilitation settings — particularly home-based — signals are contaminated by motion artifact, electrode displacement, loose contact, sweat-induced impedance changes, and electrical interference. The ML preprocessing layer must be robust to these conditions.

Key artifact sources by signal type

- ECG: Motion artifact from limb movement, electrode peel-off, muscle electrical interference (EMG contamination)
- PPG-derived HR: Motion artifact particularly during exercise and walking; pigmentation and perfusion effects on wrist PPG
- IMU: Non-wear periods (device removal), attachment variability, gravitational artifact during postural transitions

Priority methods papers for C4

- Adaptive filtering methods for ECG motion artifact (LMS, Wiener filter — IEEE Trans. Biomed. Eng.)
- Pan-Tompkins QRS detection algorithm and its successors for robust R-peak detection in noisy ECG
- Accelerometry-based ECG artifact cancellation using IMU as noise reference signal
- Signal quality index (SQI) papers: automated ECG quality assessment before feature extraction
- Non-wear detection algorithms for wrist accelerometers — GGIR software library papers

3.5 Sensor Fusion Methods

The greatest clinical insight comes not from any single sensor stream but from the combination of ECG, HR, HRV, and IMU data. Sensor fusion methods allow the system to reason about physiological state in a way that is more robust and more clinically meaningful than any modality alone.

For example, a high resting HR combined with low activity counts may indicate post-stroke fatigue, sinus tachycardia from dehydration, or cardiac decompensation — three conditions requiring entirely different clinical responses. Context from the IMU (is the patient lying down? walking?) is essential to disambiguate the HR reading.

- Multi-modal physiological state estimation papers combining ECG + accelerometry + SpO2
- Kalman filter and particle filter applications for real-time physiological state tracking
- Transfer learning approaches: pre-training on large ECG databases, fine-tuning on stroke-specific cohorts
- Federated learning papers for privacy-preserving training on distributed rehabilitation datasets
- Uncertainty quantification in wearable ML models — critical for clinical safety

Pillar D — Data Correlation & Clinical Benchmark Mapping

4. Data Correlation & Clinical Benchmark Mapping

This pillar is the connective tissue between raw wearable signals and clinical meaning. It answers the question: when the wearable shows X, what does the clinical record say about the patient's actual state? Without this translation layer, the LLM and AI agent cannot make clinically grounded inferences.

4.1 Wearable-to-Clinical Benchmark Validation Studies

Formal validation studies that correlate wearable-derived metrics with established clinical scales are the highest-priority papers in this pillar. These studies provide the empirical basis for the claim that a given wearable reading has clinical meaning — a claim that is essential for regulatory compliance and clinical acceptance of the SRMS.

- Step count vs. Barthel Index and 10-Metre Walk Test (10MWT): multiple prospective cohort studies
- Wrist accelerometry activity ratio vs. Fugl-Meyer upper extremity score (FMA-UE)
- HRV (SDNN, RMSSD) vs. modified Rankin Scale at 90 days post-stroke
- Wearable-derived sleep quality vs. post-stroke fatigue scales (Fatigue Assessment Scale, FSS)
- Resting HR variability vs. functional independence (FIM) at discharge from inpatient rehabilitation

Evidence quality note

The corpus should distinguish between correlation studies (which establish association) and prospective validation studies (which establish predictive validity). For the AI agent to act on a wearable signal as a clinical indicator, at least moderate prospective evidence should exist. The LLM can be taught to express appropriate uncertainty when acting on correlational evidence only.

4.2 Digital Biomarker Discovery

Beyond well-established metrics like HRV and step count, a growing literature identifies novel digital biomarkers from continuous wearable data that provide leading indicators of clinical change. These are particularly valuable for enabling the AI agent's predictive, rather than purely reactive, capabilities.

Emerging digital biomarkers in stroke rehabilitation

- Nocturnal HR dip (circadian rhythm restoration): Restoration of the normal nocturnal HR dip correlates with improved autonomic recovery and predicts better 6-month functional

outcome. Loss of dip is associated with poor neurological prognosis and higher recurrence risk.

- HRV circadian pattern: The normal circadian HRV rhythm (higher during sleep) is disrupted post-stroke. Its restoration trajectory predicts autonomic recovery and correlates with insular cortex lesion resolution.
- Gait variability (coefficient of variation of stride time): Elevated gait variability predicts fall risk and is more sensitive to early motor deterioration than mean gait speed.
- Activity fragmentation index: The degree to which daily activity is fragmented into short bouts (rather than sustained) predicts functional decline and poor rehabilitation engagement.
- Bilateral limb correlation coefficient: The real-time correlation between affected and unaffected limb accelerometry during bilateral tasks — a direct measure of interlimb coordination recovery.

4.3 Remote Patient Monitoring Clinical Trials

Clinical trials that specifically test telemonitoring or remote physiological monitoring in post-stroke populations provide both efficacy evidence and, critically, the clinical alert thresholds that have been prospectively validated. These are the studies that translate physiological findings into actionable clinical decision boundaries.

- STROKESTOP trial: Population-based ECG screening for AF post-stroke — alert threshold validation
- CRYSTAL AF trial: Continuous cardiac monitoring with implantable loop recorder after cryptogenic stroke
- FIND-AF and FIND-AF Randomized: Prolonged ECG monitoring duration studies; AF yield data
- StrokeCog trial: Wearable activity monitoring + cognitive assessment post-stroke
- RESTORE trial series: Remote stroke rehabilitation with wearable monitoring
- mHealth cardiac rehabilitation trials (e.g., MoMo-TCM, REMOTE-CR): Alert thresholds, patient engagement

4.4 Physiological Signal Databases for Training & Validation

The ML pipeline requires large, well-annotated physiological signal datasets for training, validation, and benchmarking. The following public databases are the primary training resources and should be formally cited and characterised in the corpus.

Database	Modalities	Relevance
PhysioNet / MIT-BIH Arrhythmia DB	12-lead ECG, annotations	Arrhythmia detection training; AF labels
PhysioNet MIMIC-IV Waveform	ICU ECG, PPG, ABP	Acute post-stroke physiological signatures

Database	Modalities	Relevance
UK Biobank ECG Substudy	12-lead ECG + accelerometry	Population norms by age; comorbidity strata
Chapman-Shaoxing 12-lead ECG	12-lead ECG, 10,000 patients	Chinese population; arrhythmia labels
PTB-XL	12-lead ECG, 21,837 records	Large-scale ML training; diverse pathologies
ISRUC-Sleep	Polysomnography + ECG	Nocturnal HR and HRV baseline norms
Stroke EHR datasets (I2B2, MIMIC-III)	Clinical text + structured data	LLM fine-tuning; clinical note understanding

Pillar E — Clinical AI Architecture & Agent Design

5. Clinical AI Architecture & Agent Design

The final pillar addresses the system design of the LLM integration and AI agent layers. This is where clinical knowledge becomes actionable. The corpus here is necessarily interdisciplinary, spanning biomedical NLP, machine learning explainability, human-computer interaction, and clinical informatics.

5.1 Clinical NLP & Medical LLM Fine-Tuning

The LLM at the heart of Layer 3 must be capable of reasoning about physiological data in clinical terms, synthesising information from the research corpus with real-time wearable outputs. General-purpose LLMs (GPT-4, Claude, Llama) have substantial medical knowledge but lack the precision required for safety-critical clinical reasoning. Specialised medical LLMs or fine-tuned adaptations are the appropriate foundation.

Key medical LLM architectures and their validation

- GatorTron (University of Florida): Trained on 82 billion words of clinical text; strong performance on clinical NLP benchmarks
- BioMedLM (Stanford CRFM): Biomedical literature-trained; strong on PubMed QA tasks
- Med-PaLM 2 (Google): Expert-level clinical question answering; MultiMedQA benchmark
- Clinical BERT / BioBERT: BERT variants fine-tuned on PubMed and MIMIC clinical notes
- MedAlpaca / Meditron-70B: Open-source medical instruction-tuned models

Retrieval-Augmented Generation (RAG) is the recommended architecture for connecting the research corpus to the LLM, as it allows the corpus to be updated without model retraining and provides source attribution for every inference — a critical feature for clinical accountability.

- RAG for clinical decision support: Lewis et al. (2020) original RAG paper; clinical RAG adaptation studies
- Vector database selection for medical text: FAISS, Pinecone, Weaviate comparison studies
- Chunking strategy for medical documents: sentence-level vs. section-level retrieval for clinical accuracy
- Hybrid retrieval: dense (semantic) + sparse (BM25) retrieval for clinical terminology handling

5.2 Explainability in Clinical AI

Regulatory bodies (FDA, EMA, MHRA) and clinical governance frameworks require that AI-generated clinical recommendations be explainable to the clinicians who act on them. A clinical AI that says 'consider anticoagulation' without stating why is not clinically acceptable. The corpus in this sub-domain provides the methodological toolkit for making every SRMS inference transparent.

- SHAP (SHapley Additive exPlanations) applied to ECG classification — which features drove the arrhythmia flag?
- Grad-CAM and saliency maps for CNN-based ECG analysis — highlighting which signal segments triggered detection
- Attention weight visualisation in Transformer-based ECG models — temporal attribution
- Counterfactual explanation methods: 'If the patient's HRV had been 10ms higher, the alert would not have triggered'
- FDA AI/ML-Based Software as a Medical Device (SaMD) Action Plan: regulatory explainability requirements
- EU AI Act high-risk AI system documentation requirements for medical devices

5.3 AI Safety & Alert Fatigue

Alert fatigue is one of the most significant safety risks in clinical decision support systems. When a monitoring system generates too many alerts — particularly false positives — clinicians rapidly develop alert fatigue and begin ignoring or overriding alerts, including true positives. The SRMS must be designed from the outset with alert burden as a primary engineering constraint.

Design principle

The literature consistently demonstrates that the optimal alert specificity for wearable cardiac monitoring in non-ICU settings is >90%, even at some cost to sensitivity. For the SRMS, a tiered alert architecture is recommended: automated escalation thresholds for high-specificity events (e.g., sustained AF >30 minutes), advisory flags for lower-specificity findings (e.g., elevated resting HR for 48 hours), and daily summaries for trends.

- Alert fatigue in hospital information systems: Kern et al. (2013) JAMIA landmark study
- Optimising alert specificity: trade-off studies in cardiac monitoring (intensive care and telemetry settings)
- Bayesian alert thresholding: incorporating prior probability of disease in alert design
- Alarm management standards: IEC 60601-1-8 medical electrical equipment alarm systems
- Post-stroke patient-reported outcome measures (PROMs) for false alarm burden — patient experience literature

5.4 Human-AI Collaboration in Rehabilitation

The AI agent's outputs must be designed to be used by two very different audiences: clinicians with high domain expertise but limited time, and patients with limited medical literacy but high

personal investment. The corpus in this domain informs both the content and the format of agent outputs.

For the clinician interface

- Clinical decision support usability studies: alert format, information density, and workflow integration
- Physician trust in AI recommendations: calibration, uncertainty display, and override tracking
- EHR integration studies for wearable data — FHIR standards and CDS Hooks implementation
- Shared decision-making frameworks that incorporate AI-generated risk estimates

For the patient interface

- Health literacy-adapted AI communication studies in stroke and cardiac populations
- mHealth engagement and adherence in stroke survivors: what keeps patients using the app?
- Patient acceptance of AI-generated health guidance: systematic reviews (2020–2024)
- Motivational factors in rehabilitation adherence — self-efficacy, social support, and feedback design

6. Corpus Curation Methodology

6.1 Search Strategy

The following databases should be searched systematically, with searches documented for reproducibility and potential systematic review compliance.

Database	Focus	Priority Pillar(s)
PubMed / MEDLINE	Clinical and biomedical literature	A, B, D
IEEE Xplore	Signal processing and ML engineering	C, E
ACM Digital Library	HCI and clinical informatics	E
arXiv (cs.LG, eess.SP, cs.CL)	Pre-print ML and NLP research	C, E
Cochrane Library	Systematic reviews and meta-analyses	A, B
ClinicalTrials.gov	Registered trials for unpublished data	B, D
PhysioNet.org	Signal databases and challenge papers	C, D
WHO ICTRP	International trial registry	B, D

6.2 Inclusion & Exclusion Criteria

Inclusion criteria

- Human subjects studies (animal studies excluded unless the only available mechanistic data)
- Adults aged 40 and above; paediatric stroke excluded
- Publication date 2000–2025 (pre-2000 landmark studies included on named basis)
- English language primary; translated landmark studies accepted
- All study designs: RCT, cohort, case-control, cross-sectional, systematic review, clinical trial
- For signal databases: publicly available with appropriate data use agreement

Exclusion criteria

- Non-stroke cerebrovascular disease as primary focus (TIA-only studies accepted if reporting cardiac monitoring data)
- Studies reporting only surrogate endpoints without functional or clinical correlation
- Conference abstracts without full-text publication
- Studies with sample size <20 without specific methodological justification

6.3 Tagging Schema for LLM Ingestion

Each paper admitted to the corpus should be tagged with the following structured metadata to enable precise retrieval by the RAG system.

Recommended tagging fields

Pillar (A–E) | Sub-category (A1–E4) | Study type (RCT / Cohort / Review / Methods / Dataset) | Patient population (age group + setting) | Data modality (ECG / HR / HRV / IMU / Clinical) | Time point (Acute / Subacute / Chronic / All) | Clinical scale reported (NIHSS / mRS / Barthel / FMA / Other) | Key finding (1–2 sentence) | Signal-to-clinical correlation (Yes / No) | Evidence quality (High / Moderate / Low)

6.4 Recommended Corpus Size & Composition

Pillar	Recommended Papers	Priority Ratio
A — Stroke physiology & cardiac	80–100 papers	High — foundational for all layers
B — Recovery staging & benchmarks	70–90 papers	High — provides training labels
C — Wearable signal processing	90–120 papers	Highest — ML pipeline directly dependent
D — Data correlation & mapping	60–80 papers	High — validation and dataset acquisition
E — Clinical AI architecture	50–70 papers	Medium-high — system design guidance
Total	350–460 papers	Reviewed and curated from ~2,000 screened

7. Conclusions & Next Steps

This literature review establishes the five-pillar research corpus required to design, train, validate, and deploy the Stroke Rehabilitation Monitoring System. Across both target age groups — middle-aged adults (40–64) and older adults (65+) — and across both rehabilitation settings — inpatient and home-based — the corpus addresses the foundational clinical questions and provides the methodological grounding for each system layer.

Three themes emerge consistently across pillars and should be treated as design imperatives for the SRMS team.

Context dependency of physiological signals. No wearable signal — HRV, resting HR, step count, or movement asymmetry — carries clinical meaning in isolation. Its interpretation is inseparable from the patient's age, comorbidity profile, recovery phase, rehabilitation setting, and time since stroke onset. The ML pipeline and LLM must be architected to reason conditionally, not through fixed thresholds.

The gap between lab and home. The majority of validation studies for wearable metrics in stroke rehabilitation were conducted in supervised clinical settings. The real-world performance of these metrics in home environments — with non-wear periods, activity variability, and absence of direct clinical observation — is substantially less well characterised. The SRMS should treat home-derived readings with calibrated uncertainty and incorporate non-wear detection and signal quality gating as mandatory pipeline components.

Alert design is a clinical safety question. The decision of when and how to alert clinicians or patients is not a technical afterthought but a primary patient safety design decision. The corpus on alert fatigue, tiered alert architecture, and regulatory requirements (FDA SaMD guidance, IEC 60601-1-8) should inform the AI agent's alert logic from the earliest design stages, not as a post-hoc refinement.

Recommended next steps

- Commission a structured systematic search across all five pillars using the search strategy in Section 6.1
- Screen approximately 2,000 papers against inclusion/exclusion criteria; target 350–460 for full corpus
- Apply the tagging schema (Section 6.3) to all included papers using a research assistant or semi-automated NLP tool
- Acquire data use agreements for PhysioNet, UK Biobank, and MIMIC-IV Waveform datasets

- Establish a living review process — quarterly literature updates to capture the rapidly evolving wearable AI literature
- Prioritise Pillar C (signal processing) and Pillar B (clinical benchmarks) for the ML pipeline development phase

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